Listing multiples

Task A: Listing multiples

1) List the first five multiples of:

a) 8 8, 16, 24, 32, 40

b) 10 10, 20, 30, 40, 50

c) 12 12, 24, 36, 48, 60

d) 15 15, 30, 45, 60, 75

2) Alex says

30 is a multiple of 4 because
$4 \times 7.5 = 30$

Is Alex correct? Explain your answer

A multiple is the product of two integers

3) Jacob started listing the first multiples of some numbers. He then spilled ink all over his work.

Find the multiples Jacob was working out and the missing numbers.

a) Multiples of 111 : 111, 222, 333, 444, 555, …

b) Multiples of 23 : 23, 46, 69, 92, 115, 138, …

c) Multiples of 324 : 324, 648, 972, 1296, 1620, …

d) Multiples of 845 : 845, 1690, 2535, 3380, 4225, …

Task B: Using listing to find common multiples

1) Find all the **common multiples** of 7 and 5 which are less than 100.

Common multiples: 35 and 70

2) Find all the **common multiples** of 5, 8 and 10 which are less than 100.

Common multiples: 40, 80

3) Using the numbers, fill in the missing gaps. You can only use each number once.

5	6	9	10	12	12	27	27	50

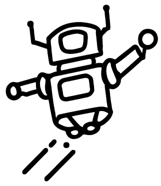
a) $\boxed{12}$ is a **common multiple** of $\boxed{6}$ and $\boxed{12}$

b) $\boxed{50}$ is a **common multiple** of $\boxed{5}$ and $\boxed{10}$

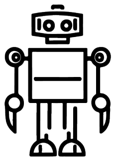
c) $\boxed{27}$ is a **common multiple** of $\boxed{27}$ and $\boxed{9}$

Task B: Using listing to find common multiples

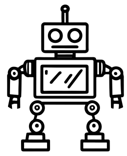
4) A path is made up from 100 steps, numbered 1 to 100.



Robot A steps on every 8th step.



Robot B steps on every 3rd step.



Robot C steps on every 6th step.

a) Which numbered steps will all three robots have stepped on?

24, 48 and 72

b) Robot D steps on every 2nd step. Will this change your answer to part a? Explain why.

No because multiples of 2 are also common multiples of 8 and 6 so it will make no difference.

Task C: Using common multiples to compare fractions

1) Using **common multiples** to compare fractions, identify the largest fraction. You must show your working out.

a) $\boxed{\frac{3}{5}}$ and $\frac{2}{7}$

$$\frac{3}{5} = \frac{6}{10} = \frac{9}{15} = \frac{12}{20} = \frac{15}{25} = \frac{18}{30} = \boxed{\frac{21}{35}}$$
$$\frac{2}{7} = \frac{4}{14} = \frac{6}{21} = \frac{8}{28} = \boxed{\frac{10}{35}}$$

b) $\frac{4}{5}$ and $\boxed{\frac{5}{6}}$

$$\frac{4}{5} = \frac{8}{10} = \frac{12}{15} = \frac{16}{20} = \frac{20}{25} = \boxed{\frac{24}{30}}$$
$$\frac{5}{6} = \frac{10}{12} = \frac{15}{18} = \frac{20}{24} = \boxed{\frac{25}{30}}$$

c) $\boxed{\frac{7}{10}}$ and $\frac{3}{8}$

$$\frac{7}{10} = \frac{14}{20} = \frac{21}{30} = \boxed{\frac{28}{40}}$$
$$\frac{3}{8} = \frac{6}{16} = \frac{9}{24} = \frac{12}{32} = \boxed{\frac{15}{40}}$$

2) Three students have the following success rates on a game.

WELL DONE SAM! YOUR SUCCESS RATE IS $\frac{2}{3}$	WELL DONE LUCAS! YOUR SUCCESS RATE IS $\frac{11}{15}$	WELL DONE ALEX! YOUR SUCCESS RATE IS $\frac{3}{5}$
$\frac{2}{3} = \frac{4}{6} = \frac{6}{9} \ldots \boxed{\frac{20}{30}}$	$\frac{11}{15} = \boxed{\frac{22}{30}} = \frac{33}{45}$	$\frac{3}{5} = \frac{6}{10} = \frac{9}{15} \ldots = \boxed{\frac{18}{30}}$

Who is the best player? You must show your working out. Lucas

State a fractional success rate in between Sam's and Alex's. $\frac{19}{30}$